

Engineering Pathways for Community College Students

MATLAB and Simulink Projects Orientation

Agenda



Introductions



Timeline



Resources



Judging, Awards, and Certificates



Questions

Your Team from MathWorks



Dr. Harish
Chintakunta
Senior Customer
Success Engineer



Dr. Esperanza
Linares
Senior Customer
Success Engineer



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Zhang
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Education
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Engineer
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Timeline

June 30th	MathWorks Challenge Orientation
July 6 - August 5th	Main phase: Teams work on the projects and MathWorks provides Office Hours
August 6th - August 7th	Wrapup phase: Teams complete their reports, and online submissions
August 10th - August 14th	Evaluation: First round of peer reviews organized by EPP to provide five nominations. Followed by EPP+MathWorks selecting 3 winners.

Overview of Classroom Challenge Projects Program

Project preparation

- Choose a project with your teams
- Establish roles and responsibilities
- Develop background knowledge and skills

Project work

- Set and work toward meeting project milestones
- Use suggested steps provided in project description as a guide
- Use provided resources for help along the way

Project Submission and Evaluation

- Prepare your project for submission
- Evaluation

General Project Resources



[How-To Videos on MATLAB YouTube](#)



GenAI Tools

[MATLAB GPT](#)

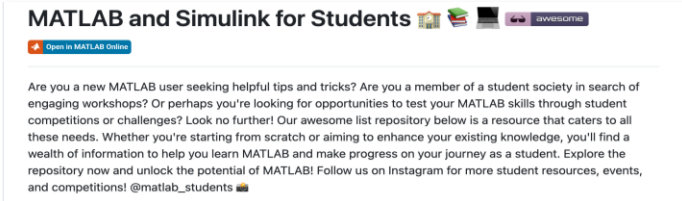
[MATLAB AI Chat Playground with Tutor](#)



[MATLAB Help Center and Documentation](#)

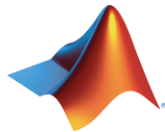


[MATLAB Student GitHub Page](#)



Weekly Office Hours:

- Harish, **Tuesdays at 2:00 pm**
- Esperanza, **Thursdays at 5:00 pm**



[MathWorks Online Training Suite](#)



[Guide for using GitHub](#)

These resources are available to you from the [Classroom Challenge Projects Hub](#)

Resources



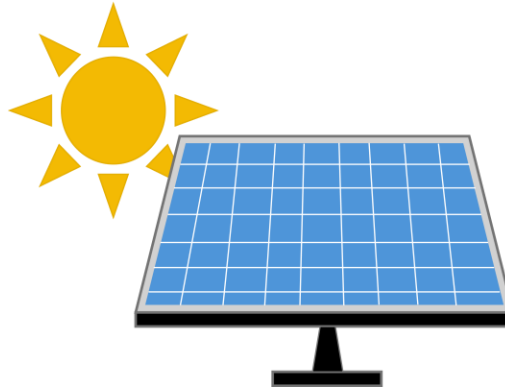
- [MATLAB Answers](#)
- [MATLAB Tech Support](#)

[Discord for Discussion Board](#)

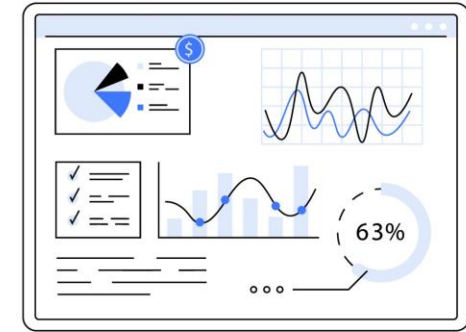
Projects Tour



Modeling and Optimizing a Battery Charging Profile



Maximizing Solar Panel Output for a Fixed Area



Data Analytics Apps for a Smart Eco Gym



Quarter-Car Suspension Modeling and Simulation

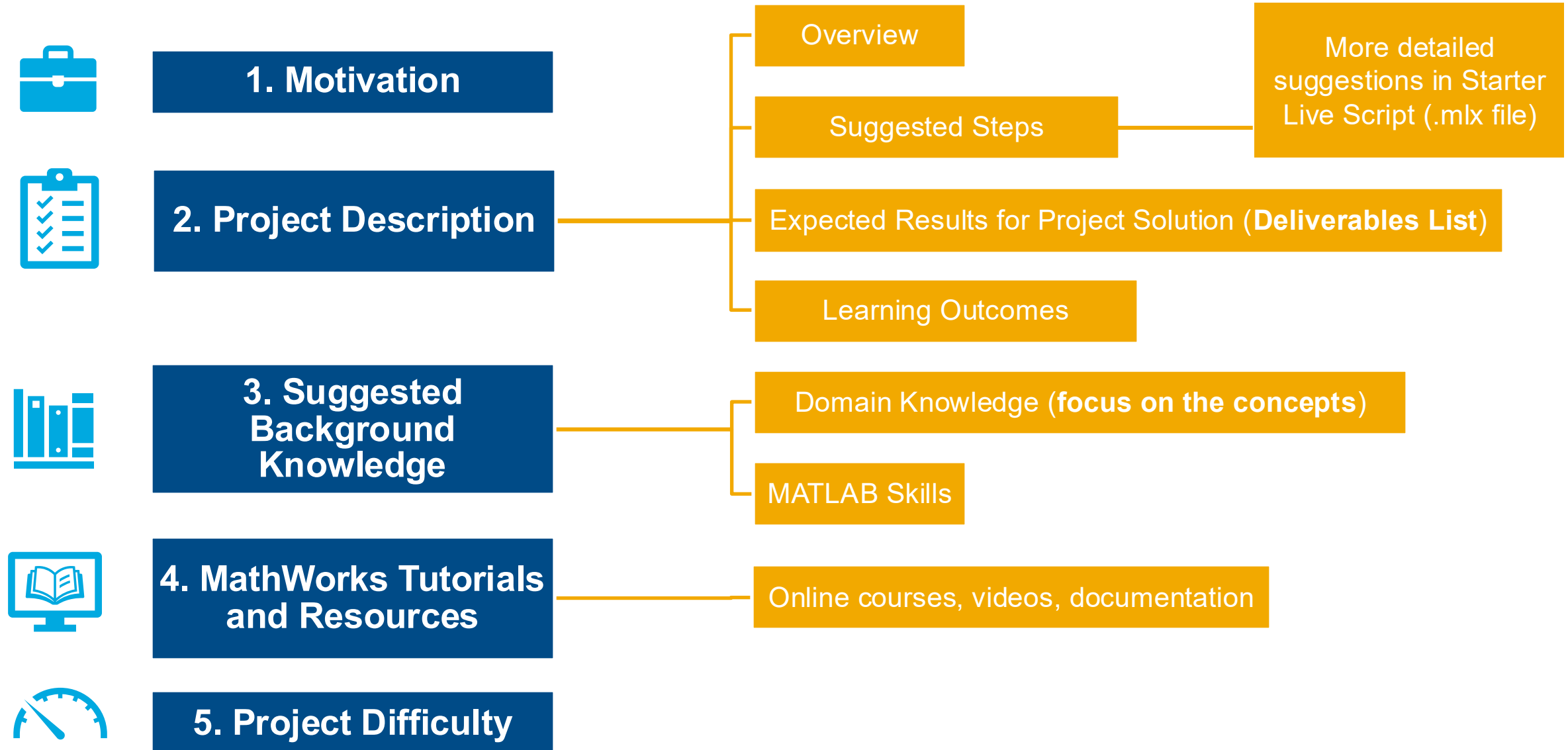


Drone Payload Structure Design and Analysis



Image-Based Defect Detection for Manufacturing Inspection

Every Project Page includes:



[Link to Drone Project Page](#)
(as an example)

Judging Awards and Certificates

3 Winner Teams



- 1st place → \$200 gift card
- 2nd place → \$100 gift card
- 3rd place → \$50 gift card



Every student will receive a digital **certificate of project completion**

EPP mentors will evaluate all projects and **nominate** up to 5 teams

- Projects due **August 7th**

Nominated projects **must** publish materials in a public GitHub repository

- [Instructions for creating GitHub repository](#)
- [Resources to get started with GitHub](#)

MathWorks + EPP will review nominated projects and select winning teams

- Winning teams announced on **August 14th**

Questions

